

## QAIROON HAIRITI, Oman

WMO: 413150

Lat: 17.253N

Lon: 54.089E

Elev: 2890

StdP: 13.22

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 50.3       | 53.2 | 13.0                          | 11.8 | 61.4 | 17.9 | 15.1 | 65.9 | 51.1                  | 58.1 | 42.4 | 58.4 | 20.9                  | 340  | 0.522 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 11.3                   | 91.6            | 62.2 | 88.2 | 64.2 | 85.7 | 65.2 | 76.1                | 82.2 | 74.9 | 80.3 | 73.9 | 78.6 | 11.5                 | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.4                            | 143.4 | 79.0 | 73.4 | 138.5 | 77.5 | 72.6 | 134.8 | 76.1 | 42.0          | 82.6 | 40.7 | 80.8 | 39.8 | 79.1 | 85.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 35.7              | 28.8 | 25.2 | DB                         | 46.8 | 100.6              | 1.7 | 2.2                                                | 45.5 | 102.2      | 44.5 | 103.4      | 43.6 | 104.7      | 42.4 | 106.2 |  |
|                   |      |      | WB                         | 35.3 | 77.2               | 2.1 | 1.3                                                | 33.8 | 78.1       | 32.6 | 78.8       | 31.4 | 79.5       | 29.8 | 80.5  |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 70.8   | 62.6  | 64.9  | 70.4  | 74.9  | 77.2  | 77.4  | 70.9  | 69.4  | 71.9  | 74.2  | 70.5  | 64.6  |
|                                                                   | DBStd       | 5.84   | 3.73  | 3.94  | 3.71  | 3.70  | 3.44  | 4.54  | 3.12  | 2.78  | 3.56  | 3.25  | 2.72  | 3.75  |
|                                                                   | HDD50       | 1      | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | HDD65       | 184    | 87    | 42    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 52    |
|                                                                   | CDD50       | 7579   | 391   | 418   | 633   | 748   | 844   | 823   | 649   | 601   | 657   | 749   | 614   | 452   |
|                                                                   | CDD65       | 2288   | 12    | 40    | 171   | 298   | 379   | 373   | 184   | 137   | 207   | 284   | 164   | 39    |
|                                                                   | CDH74       | 10979  | 12    | 83    | 649   | 1821  | 2609  | 2577  | 281   | 142   | 804   | 1529  | 428   | 44    |
|                                                                   | CDH80       | 2937   | 0     | 4     | 87    | 449   | 791   | 994   | 96    | 32    | 177   | 281   | 26    | 0     |
| Wind                                                              | WSAvg       | 14.0   | 13.6  | 14.1  | 13.8  | 14.6  | 14.2  | 14.6  | 17.8  | 16.7  | 12.4  | 10.4  | 11.4  | 14.5  |
| Precipitation                                                     | PrecAvg     | 4.5    | 0.0   | 0.0   | 0.1   | 0.2   | 0.4   | 0.3   | 1.5   | 1.3   | 0.3   | 0.1   | 0.2   | 0.0   |
|                                                                   | PrecMax     | 11.3   | 0.2   | 0.3   | 1.2   | 2.6   | 6.1   | 1.4   | 4.3   | 3.5   | 3.0   | 0.9   | 3.0   | 1.0   |
|                                                                   | PrecMin     | 0.9    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                   | PrecStd     | 2.7    | 0.0   | 0.0   | 0.3   | 0.5   | 1.2   | 0.4   | 1.1   | 0.9   | 0.6   | 0.2   | 0.6   | 0.2   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 75.8  | 79.4  | 85.8  | 89.9  | 95.8  | 98.9  | 89.6  | 85.1  | 89.2  | 87.5  | 82.7  | 77.4  |
|                                                                   |             | MCWB   | 57.2  | 57.9  | 59.5  | 60.1  | 61.4  | 62.0  | 77.4  | 72.2  | 63.7  | 60.9  | 59.8  | 59.7  |
|                                                                   | 2%          | DB     | 72.7  | 76.4  | 82.3  | 87.2  | 90.1  | 93.5  | 80.4  | 77.3  | 84.6  | 85.2  | 80.2  | 75.0  |
|                                                                   |             | MCWB   | 57.8  | 57.3  | 58.2  | 60.5  | 64.4  | 61.2  | 73.8  | 71.3  | 63.3  | 63.0  | 60.3  | 59.1  |
|                                                                   | 5%          | DB     | 70.6  | 74.1  | 79.8  | 84.9  | 86.7  | 89.2  | 75.8  | 73.7  | 81.4  | 83.1  | 78.2  | 73.2  |
|                                                                   |             | MCWB   | 58.0  | 57.6  | 57.9  | 61.4  | 66.7  | 64.7  | 72.5  | 71.5  | 66.2  | 63.5  | 60.7  | 58.6  |
|                                                                   | 10%         | DB     | 68.9  | 71.8  | 77.7  | 82.6  | 83.9  | 85.5  | 73.2  | 71.8  | 78.3  | 81.1  | 76.4  | 71.2  |
|                                                                   |             | MCWB   | 57.5  | 57.6  | 57.7  | 62.8  | 68.2  | 68.8  | 71.6  | 70.7  | 68.1  | 63.4  | 60.9  | 58.6  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 64.4  | 67.2  | 69.3  | 72.7  | 76.1  | 77.3  | 75.9  | 75.1  | 75.8  | 80.4  | 74.5  | 69.5  |
|                                                                   |             | MCDB   | 69.0  | 71.5  | 75.4  | 79.2  | 83.9  | 85.4  | 81.9  | 80.9  | 80.2  | 83.3  | 76.9  | 71.7  |
|                                                                   | 2%          | WB     | 62.8  | 64.9  | 67.8  | 71.4  | 74.5  | 76.1  | 74.1  | 73.0  | 73.6  | 74.2  | 69.8  | 66.0  |
|                                                                   |             | MCDB   | 67.3  | 69.7  | 72.4  | 77.3  | 81.3  | 82.7  | 77.6  | 75.6  | 78.3  | 78.0  | 73.2  | 68.8  |
|                                                                   | 5%          | WB     | 61.7  | 63.4  | 66.8  | 70.5  | 73.6  | 75.2  | 72.7  | 71.8  | 72.2  | 70.8  | 67.5  | 63.8  |
|                                                                   |             | MCDB   | 66.0  | 67.9  | 71.0  | 76.0  | 79.7  | 80.8  | 74.5  | 73.1  | 76.0  | 76.4  | 71.4  | 67.9  |
|                                                                   | 10%         | WB     | 60.8  | 62.1  | 65.6  | 69.6  | 72.8  | 74.3  | 71.9  | 70.8  | 71.1  | 69.2  | 65.8  | 62.2  |
|                                                                   |             | MCDB   | 64.8  | 66.5  | 69.9  | 74.9  | 78.4  | 79.1  | 72.8  | 71.6  | 74.1  | 74.3  | 69.9  | 66.8  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.0  | 13.2  | 13.3  | 13.4  | 13.3  | 11.3  | 3.9   | 4.1   | 10.0  | 13.9  | 12.9  | 13.0  |
|                                                                   |             | MCDBR  | 14.2  | 15.0  | 15.3  | 15.8  | 18.0  | 20.0  | 10.7  | 9.3   | 16.0  | 15.5  | 14.2  | 14.2  |
|                                                                   |             | MCWBR  | 7.6   | 8.7   | 10.0  | 11.0  | 10.6  | 12.8  | 4.7   | 6.0   | 12.4  | 10.9  | 8.8   | 7.5   |
|                                                                   | 5% WB       | MCDBR  | 11.9  | 12.1  | 12.1  | 12.7  | 13.3  | 12.5  | 7.0   | 7.3   | 12.0  | 14.4  | 12.7  | 12.7  |
|                                                                   |             | MCWBR  | 6.1   | 6.6   | 6.9   | 6.7   | 5.7   | 6.8   | 4.0   | 5.1   | 9.3   | 10.9  | 9.1   | 8.4   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.364 | 0.383 | 0.428 | 0.457 | 0.534 | 0.696 | 0.744 | 0.667 | 0.529 | 0.397 | 0.379 | 0.365 |
|                                                                   | taud        |        | 2.256 | 2.184 | 2.047 | 1.995 | 1.809 | 1.508 | 1.465 | 1.600 | 1.843 | 2.191 | 2.236 | 2.253 |
|                                                                   | Ebn at Noon |        | 291   | 291   | 280   | 272   | 248   | 209   | 200   | 217   | 249   | 283   | 284   | 287   |
|                                                                   | Edh at Noon |        | 41    | 46    | 55    | 58    | 69    | 93    | 97    | 85    | 67    | 45    | 42    | 40    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 1720  | 1971  | 2189  | 2321  | 2340  | 2134  | 1866  | 1905  | 2049  | 2028  | 1764  | 1641  |
|                                                                   | RadStd      |        | 46    | 45    | 75    | 61    | 69    | 95    | 91    | 54    | 64    | 49    | 59    | 51    |

## Historical Trends

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (4 neighbors) | +0.27 | N/A     | -2.74  | N/A   | N/A     | -0.54 | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page